

Practical Linux Administration & NVIDIA GPU Management

Enterprise Systems, Virtualization & High-Performance Computing

2-Day Training

Dual-Distribution: **Ubuntu 24.04 LTS & Rocky Linux 9**

Datacenter Compute: **NVIDIA GPU Administration & Multi-Instance GPU (MIG)**

Daily Schedule: 10:00 – 16:00 | Lunch Break: 13:00 – 14:00

JUL

17

2-Day Training Timetable & Syllabus

Day	Time	Module	Core Topics & Practical Focus
Day 1	10:00 – 11:30	Module 1	Terminal Basics: UNIX philosophy, shell prompt, streams, paths, nano.
	11:30 – 13:00	Module 2	File Structure & Security: FHS tree, user/group models, sudo, octal permissions.
	13:00 – 14:00	LUNCH	<i>Mid-day Break</i>
	14:00 – 15:00	Module 3	Package Management: APT (<code>.deb</code>) vs DNF (<code>.rpm</code>), repo architecture, updates.
	15:00 – 16:00	Module 4	Storage & Filesystems: Block devices, <code>df</code> / <code>du</code> , loopback disks, ext4 vs XFS.
Day 2	10:00 – 11:30	Module 5	Processes & Systemd: Process model, signals, PID 1, systemd unit lifecycle.
	11:30 – 13:00	Module 6	Networking & SSH: TCP/IP sockets, ports, asymmetric cryptography, Ed25519.
	13:00 – 14:00	LUNCH	<i>Mid-day Break</i>
	14:00 – 15:00	Module 7	Security & Logs: Centralized logging, <code>systemd-journald</code> , <code>tail -f</code> , filtering.
	15:00 – 16:00	Module 8	NVIDIA GPU & MIG: GPU architecture, <code>nvidia-smi</code> metrics, hardware MIG slicing.

Lab Architecture: Dual-Distro + Datacenter GPU

Local Virtualization (Canonical Multipass)

- **Ubuntu 24.04 LTS (`ubuntu`)**
 - Standard in modern cloud, AI developer tooling, and containers.
 - Package manager: `apt` / `dpkg` .
 - Default filesystem: `ext4` .
- **Rocky Linux 9 (`rockylinux`)**
 - RHEL 9 binary-compatible enterprise standard.
 - Package manager: `dnf` / `rpm` .
 - Default filesystem: `xfs` .
- Isolated hypervisor bridge network.

Remote AI/HPC Server

- **Bare-Metal NVIDIA GPU Server**
 - Enterprise Datacenter GPU (A100 / H200).
 - Production NVIDIA Linux Kernel Drivers & NVML.
 - Silicon-level Multi-Instance GPU (MIG) capability.
 - Dedicated participant SSH access for Module 8.
- Simulates real-world production AI cluster nodes.

[!NOTE]

Learning both the Debian and RHEL ecosystems equips administrators to handle virtually any corporate Linux server environment.

DAY 1: Foundation & Daily File Management

Module 1: Terminal Basics in Multipass

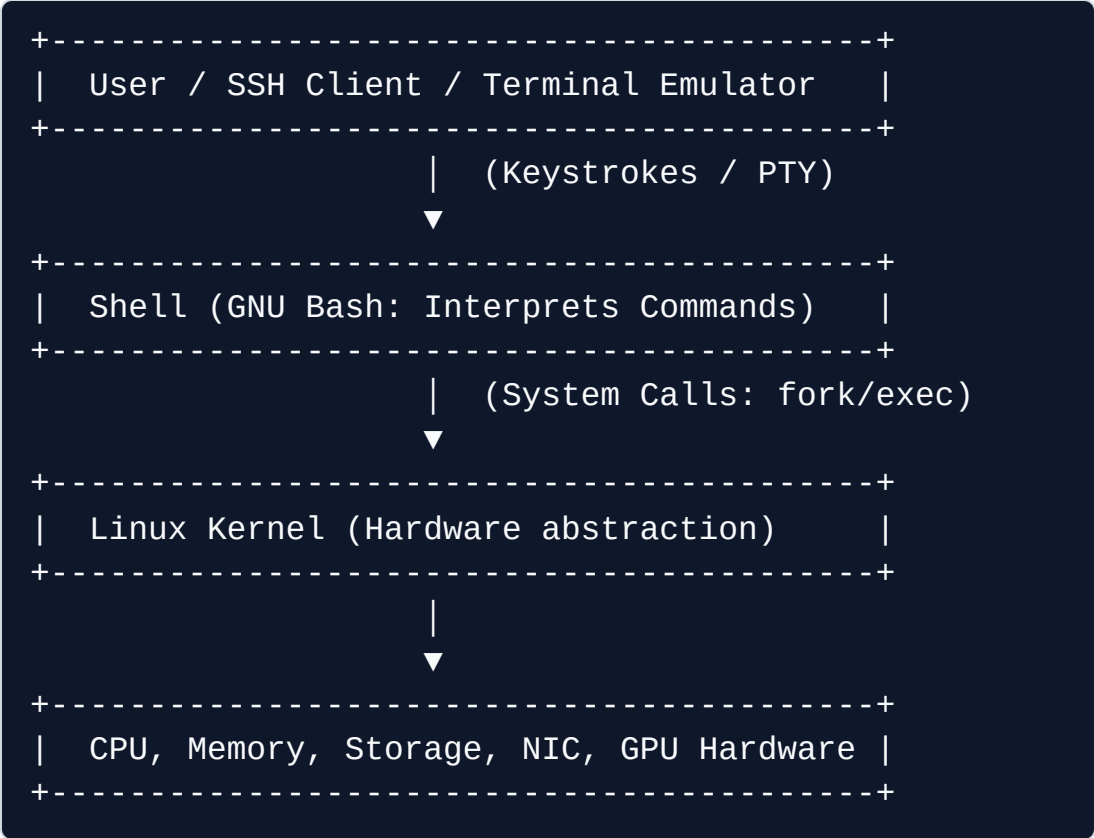
The UNIX Philosophy • Terminal Architecture • Shell Streams • Navigation • Nano

The UNIX Philosophy & Shell Architecture

Core UNIX Principles (McIlroy & Ritchie)

- 1. **Do one thing and do it well:** Modular utilities.
- 2. **Universal text streams:** Output of one tool becomes input to another.
- 3. **Flat text files:** Human-readable, versionable configurations.
- 4. **Everything is a file:** Storage, hardware devices, and network sockets are accessed through the VFS.

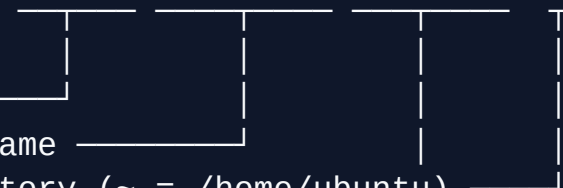
System Interaction Architecture



Anatomy of the Terminal Prompt

In modern Linux distributions, the shell prompt displays four essential operational indicators:

```
ubuntu@server-01:~/projects$
```



1. Current User
2. Machine Hostname
3. Working Directory (~ = /home/ubuntu)
4. Privilege Level (\$ = Standard User, # = Root/Admin)

Standard User (\$)

- Sandboxed execution permissions.
- Restricted to personal home directory (~).
- Cannot alter system configuration or delete core OS files without explicit privilege elevation (`sudo`).

Superuser / Root (#)

- Unrestricted kernel-level capabilities.
- Bypasses standard POSIX permission checks.
- **Enterprise Best Practice:** Avoid permanent root logins. Use `sudo` to maintain accountability and audit logs.

Path Traversal: Absolute vs Relative Paths

All directories in Linux branch from a single inverted tree rooted at `/`.

Absolute Paths

- Always start with the root directory (`/`).
- Point to the exact same file regardless of where you currently are.
- Mandatory for automated scripts, crontabs, and systemd units.
- **Examples:**
 - `/etc/nginx/nginx.conf`
 - `/var/log/syslog`
 - `/home/ubuntu/training/notes.txt`

Relative Paths

- Defined relative to the current working directory (`pwd`).
- Shorter to type during interactive terminal work.
- Uses navigational shortcuts:
 - `.` : Current directory (`./run.sh`)
 - `..` : Immediate parent directory (`cd ..`)
 - `~` : Current user's home directory (`~/notes.txt`)
 - `-` : Previous working directory (`cd -`)

```
# Equivalent ways to reach the same directory:
cd /home/ubuntu/training/day1      # Absolute path
cd ~/training/day1                 # User home relative
cd ../day1                         # Relative from /home/ubuntu/other
```

Command Syntax & The Three Standard Streams

Linux commands follow a standard format: `command [-flags] [arguments]`

Processes communicate with the environment using three standard POSIX file descriptors:

standard input (stdin)

- **FD 0**
- Input data stream.
- Defaults to keyboard.
- Redirect with `<`:

```
mysql db < dump.sql
```

standard output (stdout)

- **FD 1**
- Normal output stream.
- Defaults to screen.
- Redirect with `>` (overwrite) or `>>` (append):

```
ls -la > list.txt
```

standard error (stderr)

- **FD 2**
- Diagnostic & error logs.
- Defaults to screen.
- Redirect with `2>` or merge with stdout using `2>&1`:

```
cmd > log 2>&1
```

The UNIX Pipeline (|)

Connects the **stdout** of one process directly to the **stdin** of another via kernel memory:

```
cat /var/log/syslog | grep "Failed password" | awk '{print $11}' | sort | uniq -c
```



Headless Editing: GNU Nano Essentials

Most enterprise Linux servers operate without a Graphical User Interface (headless). Text configuration files must be inspected and edited directly via the CLI.

GNU nano Characteristics

- Pre-installed across Debian/Ubuntu and RHEL/Rocky.
- Modeless editor (keystrokes directly insert text).
- Shortcut key conventions displayed at the bottom:
 - `^` indicates the **Ctrl** key.
 - `M-` indicates the **Alt / Meta** key.

Core Shortcut Cheat Sheet

- `Ctrl + O` : **WriteOut** (Save changes to disk).
- `Ctrl + X` : **Exit** (Prompts to save if modified).
- `Ctrl + W` : **Where Is** (Search text).
- `Ctrl + K` : **Cut Line** (Delete current line).
- `Ctrl + U` : **Uncut Line** (Paste cut line).
- `Ctrl + _` : **Jump to line number**.

[!TIP]

Hands-On Lab 1 Milestone: Launch the Multipass Ubuntu shell (`multipass shell ubuntu`), practice path navigation, inspect hidden dotfiles with `ls -la`, and create configuration files using `nano`.

DAY 1: Foundation & Daily File Management

Module 2: File Structure, Users & Permissions

Filesystem Hierarchy Standard (FHS) • Users & Groups • Sudo Architecture • Permissions

Filesystem Hierarchy Standard (FHS)

Linux organizes all storage, devices, and filesystems into a single unified directory tree under `/`:

```
/ (Root Directory)
├── bin -> usr/bin      # Fundamental user command binaries (ls, cp, bash)
├── boot                # Linux kernel images (vmlinuz), initramfs, GRUB bootloader
├── dev                 # Device nodes representing hardware (/dev/sda, /dev/null, /dev/nvidia*)
├── etc                 # Host-specific system-wide text configurations (nginx, ssh, passwd)
├── home                # User personal workspaces (/home/ubuntu, /home/alex)
├── lib -> usr/lib       # Shared system libraries (.so) required by binaries
├── media / mnt         # Mount points for removable media and manual filesystem mounts
├── opt                 # Add-on third-party software packages
├── proc                # Virtual pseudo-filesystem exposing live kernel & process state
├── root                # Home directory of the root superuser (NOT /home/root!)
├── sys                 # Virtual pseudo-filesystem exposing device drivers & hardware hierarchy
├── tmp                 # Ephemeral temporary scratch space (wiped on reboot)
├── usr                 # Secondary hierarchy for user utilities, libraries, and docs
└── var                 # Variable operational data: logs (/var/log), spool, caches, web roots
```

Multi-User Architecture & Security Boundaries

Linux isolates users and daemons to protect the operating system from unauthorized changes or security exploits.

1. Account Tiers

- **Root (UID 0):** Superuser with unrestricted access.
- **System Accounts (UID 1–999):** Unprivileged service daemons (`nginx` , `sshd`). No login shell.
- **Regular Users (UID 1000+):** Human administrators and developers.

2. Core Auth Databases

- `/etc/passwd` : Publicly readable user catalog (Username, UID, GID, Home, Shell).
- `/etc/shadow` : Restricted file (mode 600/000) containing salted password hashes.
- `/etc/group` : Defines group names and member lists.

3. User Commands

- `useradd -m -s /bin/bash <user>` : Create user with home directory and shell.
- `passwd <user>` : Set or update password.
- `usermod -aG <group> <user>` : Append user to secondary group.
- `id <user>` : Inspect active UID, GID, and groups.

Privilege Escalation: The `sudo` Subsystem

`sudo` (*Superuser Do*) allows authorized users to execute specific commands with root privileges while logging actions.

Why Avoid Shared Root Logins?

- Shared root passwords prevent auditing—impossible to identify *who* made changes.
- Password revocation requires changing credentials across all hosts.
- `sudo` authenticates using the **user's own password**.
- All administrative actions are recorded in security logs (`/var/log/auth.log` or `/var/log/secure`).

Distro Differences: Sudo Groups

Ubuntu and Rocky Linux use different default administrative groups defined in `/etc/sudoers`:

- **Ubuntu / Debian Family:**

```
sudo usermod -aG sudo alex
```

- **Rocky Linux / RHEL Family:**

```
sudo usermod -aG wheel alex
```

(Derived from the historic Unix term "wheel" group)

[!WARNING]

Always edit `/etc/sudoers` using `sudo visudo`. It performs syntax validation before saving, preventing syntax errors from locking administrators out of the system.

POSIX Permission Architecture: Decoded

Every file and directory in Linux has an owner (**User**), an owning **Group**, and a 9-bit permission mode string:

```
- r w x r - x r - - 1 ubuntu staff 4096 Sep 7 10:00 deploy.sh
```

File Type ——— T ————
User (Owner) ——— U ————
Group Members ——— G ————
Others (World) ——— O ————

———— Group Owner
———— User Owner

Permissions on Files

- **Read (r = 4):** View file contents (`cat` , `nano`).
- **Write (w = 2):** Modify or overwrite file contents.
- **Execute (x = 1):** Run the file as a program or script.

Permissions on Directories

- **Read (r = 4):** List filenames inside the folder (ls).
- **Write (w = 2):** Create, rename, or delete files inside!
- **Execute (x = 1):** Enter / cd into folder or access inodes inside!

Permissions are calculated by summing the octal values of each 3-bit triplet:

Octal Value = Read(4) + Write(2) + Execute(1)

Triplet	Binary	Octal	Access Level
<code>rwX</code>	<code>111</code>	7	Full Read, Write & Execute
<code>rw-</code>	<code>110</code>	6	Read & Write (Normal data file)
<code>r-X</code>	<code>101</code>	5	Read & Execute (Programs & dirs)
<code>r--</code>	<code>100</code>	4	Read-Only
<code>---</code>	<code>000</code>	0	No access permissions

```
# Applying permissions and ownership:
chmod 755 script.sh      # rwxr-xr-x
chmod 644 config.yaml    # rw-r--r--
chmod 600 id_ed25519     # rw-----
chown alex:developers project/
```

Common Enterprise Modes

- **Mode 755 (`rwxr-xr-x`)**: Executable scripts, public directories, web roots.
- **Mode 644 (`rw-r--r--`)**: Configuration files, static assets, documents.
- **Mode 600 (`rw-----`)**: Sensitive credentials, private SSH keys, TLS certificates.
- **Mode 700 (`rwx-----`)**: Private user home directories or `~/.ssh`.

[!TIP]
Hands-On Lab 2 Milestone: Create user `alex`, configure sudo/wheel privileges, and apply octal permissions to protect sensitive files.

LUNCH BREAK (13:00 – 14:00)

Morning Wrap-up & Afternoon Preview

Mid-day Break

Afternoon Sessions:

- **Module 3 (14:00 – 15:00):** Package Management (`apt` vs `dnf` , Repositories, Updates)
- **Module 4 (15:00 – 16:00):** Storage & Virtual Filesystem Labs (Block Devices, Loopback, ext4 vs XFS)

DAY 1: Foundation & Daily File Management

Module 3: Package Management (APT vs DNF)

Package Architecture • Repositories • Metadata Indexing • APT vs DNF Rosetta Stone

Why Package Managers? Overcoming Dependency Hell

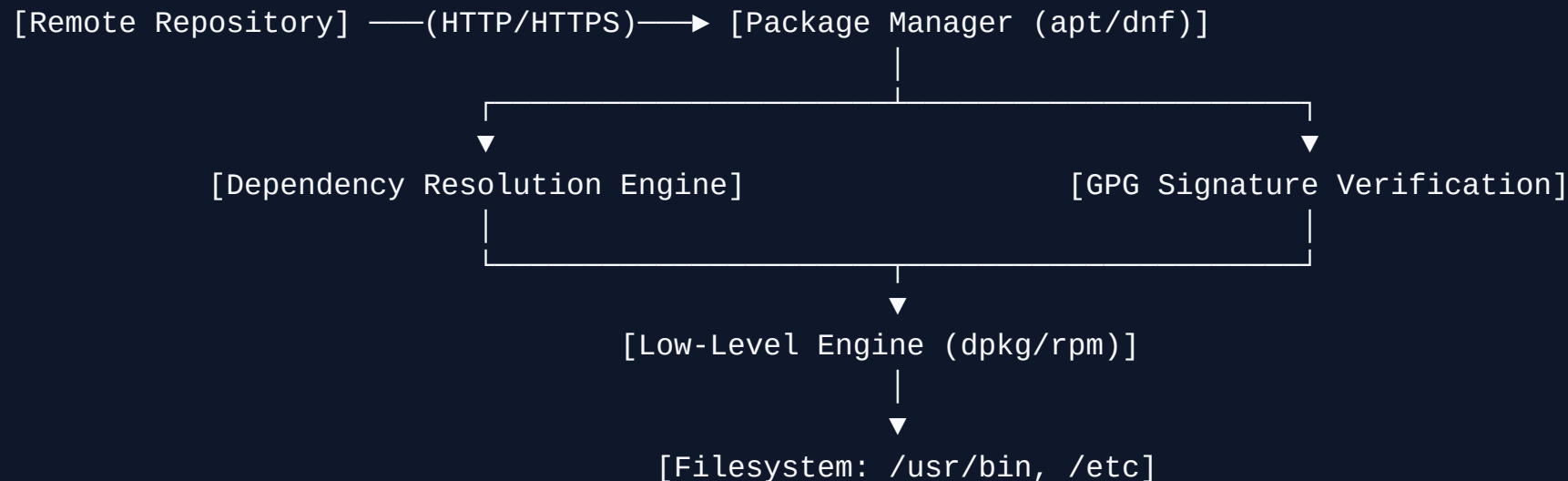
Modern package managers automate downloading, verifying, configuring, and updating software.

❌ Manual Source Compilation

- Compiling from source required manual dependency resolution (`make` , `gcc`).
- Updating libraries frequently broke existing software.
- No central catalog for cleanly uninstalling or verifying software integrity.

✅ Modern Package Management

- Pre-compiled binaries targeted to CPU architectures (`x86_64` , `aarch64`).
- Automatic dependency graph calculation.
- Cryptographic verification via GPG repository signatures.
- Clean software lifecycle: install, upgrade, rollback, remove.



Debian/Ubuntu vs RHEL/Rocky Architecture

● Ubuntu / Debian (apt + dpkg)

- **Package Format:** `.deb`
- **Low-Level Tool:** `dpkg` (installs local `.deb` files; does not fetch remote dependencies).
- **High-Level Tool:** `apt` / `apt-get` (fetches from mirrors, calculates dependency tree).
- **Configuration:** `/etc/apt/sources.list` and `/etc/apt/sources.list.d/`
- **Package Cache:** `/var/cache/apt/archives/`

● Rocky Linux / RHEL (dnf + rpm)

- **Package Format:** `.rpm` (Red Hat Package Manager)
- **Low-Level Tool:** `rpm` (unpacks payloads, manages RPM database).
- **High-Level Tool:** `dnf` (*Dandified YUM*, SAT-based dependency solver).
- **Configuration:** `/etc/yum.repos.d/*.repo`
- **Package Cache:** `/var/cache/dnf/`

[!IMPORTANT]

Key Concept: `apt update` does **not** upgrade installed packages! It only downloads the latest index metadata from upstream mirrors. You must follow with `apt upgrade` to apply updates.

Package Management: Side-by-Side Rosetta Stone

Operational Task	Ubuntu 24.04 (apt)	Rocky Linux 9 (dnf)
Refresh repository metadata	<code>sudo apt update</code>	<code>sudo dnf check-update</code>
Upgrade installed packages	<code>sudo apt upgrade -y</code>	<code>sudo dnf upgrade -y</code>
Search for available package	<code>apt search <package></code>	<code>dnf search <package></code>
Display package details	<code>apt show <package></code>	<code>dnf info <package></code>
Install package(s)	<code>sudo apt install -y <pkg></code>	<code>sudo dnf install -y <pkg></code>
Remove package (keep configs)	<code>sudo apt remove <pkg></code>	<code>sudo dnf remove -y <pkg></code>
Purge package + configs	<code>sudo apt purge <pkg></code>	<code>sudo dnf remove -y <pkg></code>
Remove unused dependencies	<code>sudo apt autoremove -y</code>	<code>sudo dnf autoremove -y</code>
Find which package owns file	<code>dpkg -S /path/to/file</code>	<code>rpm -qf /path/to/file</code>
Clean local package cache	<code>sudo apt clean</code>	<code>sudo dnf clean all</code>

[!TIP]
Hands-On Lab 3 Milestone: Install and verify `nginx` and `htop` simultaneously on Ubuntu (`apt`) and Rocky Linux (`dnf`).

DAY 1: Foundation & Daily File Management

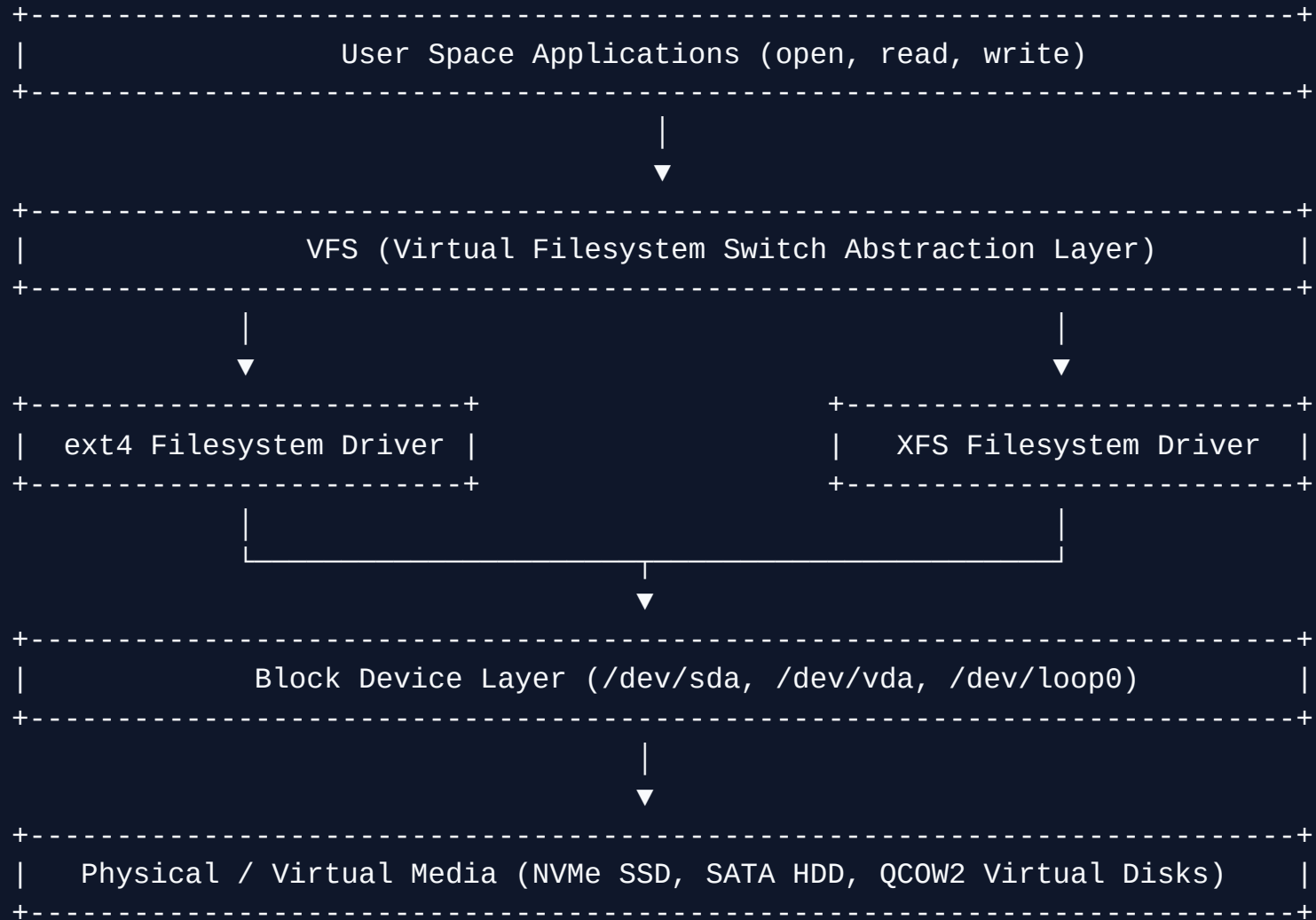
Module 4: Storage & Virtual Filesystem Labs

Storage Stack • Telemetry (df vs du) • Loopback Disks • ext4 vs XFS • Mount Lifecycle



The Linux Storage Architecture Stack

Linux uses the **Virtual Filesystem Switch (VFS)** to present heterogeneous storage hardware through a uniform file interface:



Storage Telemetry: `df` vs `du`

Understanding the difference between filesystem-level and file-level metrics is critical for disk troubleshooting:

`df -h` (Disk Free)

- Queries filesystem metadata from the **superblock**.
- Instantaneous output regardless of filesystem size.
- Reports capacity in human-readable units (`-h`).
- Displays mount points, total blocks, used blocks, and available inodes (`df -i`).

`du -sh` (Disk Usage)

- Recursively traverses directories file by file.
- Calculates actual disk space consumed by files.
- Slower on large directory trees.
- **Top space consumer command:**

```
sudo du -sh /var/* | sort -h
```

[!CAUTION]

Deleted Open File Phenomenon: If an application holds a file open when an admin deletes it with `rm`, the directory entry is removed, but disk blocks stay allocated until the process terminates! `df` reports full disk while `du` shows empty. Use `sudo lsof +L1` to identify unlinked open files.

Loopback Block Devices & Filesystems: ext4 vs XFS

A **loopback device** (`/dev/loop*`) allows mounting a regular file on disk as if it were a physical block device:

Virtual Disk Creation Pipeline

- 1. **Allocate disk image with `dd` :**
`dd if=/dev/zero of=disk.img bs=1M count=100`
- 2. **Format filesystem structure:**
 - Ubuntu standard: `mkfs.ext4 disk.img`
 - Rocky Linux standard: `mkfs.xfs disk.img`
- 3. **Mount to VFS directory:**
`mount -o loop disk.img /mnt/virtual_storage`

Filesystem Comparison

Feature	ext4 (Fourth Extended)	XFS (High-Performance)
Default OS	Ubuntu / Debian	Rocky / RHEL
Primary Focus	General-purpose stability	High-throughput concurrency
Max File Size	16 TiB	8 EiB
Shrink Support	Yes (<code>resize2fs</code>)	No (Cannot shrink XFS)
Metadata	Block groups	Allocation groups

Mounting & The Filesystem Table (/etc/fstab)

Mounting attaches a formatted storage device to a designated directory (mount point):

Mount Operations

```
# Create mount directory:
sudo mkdir -p /mnt/virtual_storage

# Mount loopback image:
sudo mount -o loop /var/virtual_disk.img /mnt/virtual_storage

# Write data into the mounted disk:
echo "UKM Training" | sudo tee /mnt/virtual_storage/test.txt

# Inspect active mounts:
mount | grep virtual_storage

# Clean unmount (flushes cached buffers to disk):
sudo umount /mnt/virtual_storage
```

/etc/fstab (Static Mount Table)

Manual `mount` commands do not survive reboot.

Permanent mounts must be declared in `/etc/fstab` with 6 fields:

# <device>	<mount point>	<type>	<options>	<dump>	<pass>
UUID=3a8f...-01	/data	xfs	defaults	0	2

1. Device Identifier (UUID or `/dev/sdX`)
2. Target Mount Point
3. Filesystem Type (`ext4` , `xfs` , `nfs`)
4. Mount Options (`defaults` , `ro` , `noexec`)
5. Dump backup utility flag (`0` to disable)
6. Filesystem check order (`1` for root, `2` for others)

[!TIP]

Hands-On Lab 4 Milestone: Create a 100MB loopback file, format it with `ext4` / `xfs` , and mount it to `/mnt/virtual_storage` .

🏁 Day 1 Wrap-up & Core Competencies Mastered

✅ Day 1 Milestones Completed

1. Headless terminal navigation, standard streams, and pipelines.
2. File operations and safe CLI editing with `nano`.
3. Navigating the Linux FHS directory layout.
4. User management, sudo security, and octal permissions (`chmod 755/644/600`).
5. Dual-distro package management (`apt` vs `dnf`).
6. Disk telemetry with `df` / `du`, loopback virtual devices, and mounting.

🚀 Day 2 Roadmap

- **Module 5:** Process management, kill signals, and systemd service control.
- **Module 6:** Network diagnostics, port binding, and Ed25519 SSH keys.
- **Module 7:** System observability with `systemd-journald`, `tail -f`, and `grep`.
- **Module 8:** Enterprise NVIDIA GPU server telemetry and Multi-Instance GPU (MIG) slicing!

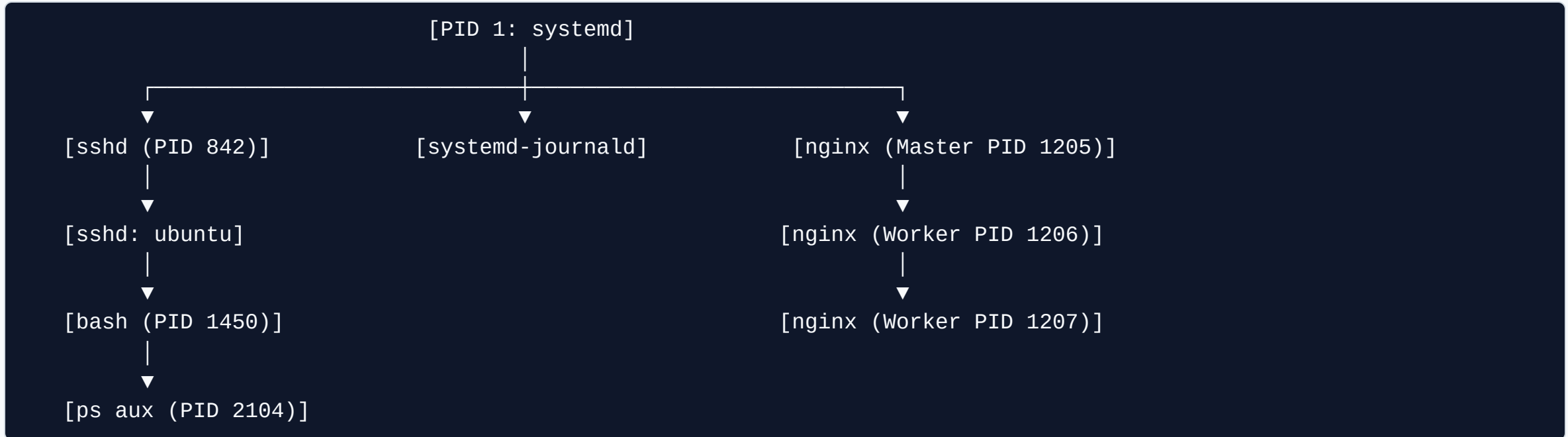
DAY 2: Services, Networking, Security & GPU Admin

Module 5: Processes & Systemd Services

Process Model • Lifecycle States • Kill Signals • Systemd Architecture • Unit Management

The Linux Process Model & Process Tree

A **process** is an actively executing program in memory, with its own address space, open file descriptors, credentials, and CPU execution context:



- **PID (Process ID):** Unique identifier assigned by the Linux kernel.
- **PPID (Parent Process ID):** The process that invoked the child via `fork()`.
- **PID 1 (`systemd`):** The root ancestor of all user-space processes on modern Linux.

Process Lifecycle & Execution States

Processes cycle through five main execution states tracked by the kernel scheduler:

Process States (Reported by `ps` / `top`)

- **R (Running / Runnable):** Executing on CPU or waiting in the active run queue.
- **S (Interruptible Sleep):** Waiting for an event or I/O. Responds to signals.
- **D (Uninterruptible Sleep):** Blocked waiting directly on hardware I/O. Cannot be killed, even with `kill -9`.
- **T (Stopped / Traced):** Suspended by user (`ctrl + Z`) or an attached debugger.
- **Z (Zombie / Defunct):** Terminated, but parent process has not yet read exit status via `waitpid()`. Consumes no RAM, only a PID table slot.

Monitoring Commands

- `ps aux` : Comprehensive snapshot of running processes.
- `ps -ef` : Traditional UNIX format showing PPID relationships.
- `top` / `htop` : Real-time interactive process dashboard.
- `pgrep <name>` : Search active PIDs by process name.
- `pstree -p` : Visual process hierarchy tree showing PIDs.

Inter-Process Signals: Terminating & Reloading

Signals are asynchronous messages sent to a process to notify it of system events or request state changes:

Signal	Name	Behavior & Enterprise Usage
1	SIGHUP	<i>Hangup.</i> Modern daemons reload configuration without dropping active client connections.
2	SIGINT	<i>Interrupt.</i> Generated by typing <code>ctrl + c</code> . Requests polite termination.
9	SIGKILL	<i>Kill.</i> Handled directly by kernel. Cannot be caught or ignored. Immediate teardown.
15	SIGTERM	<i>Terminate.</i> Default signal sent by <code>kill <PID></code> . Allows graceful cleanup and buffer flushing.
18 / 19	CONT / STOP	Resume execution / Suspend process.

Best Practice: Escalation Pattern

Always attempt graceful termination first:

```
# 1. Polite termination request:
sudo kill -15 <PID>

# 2. Wait 5-10 seconds for graceful cleanup...

# 3. Force termination only if hung:
sudo kill -9 <PID>

# Terminate by process name:
sudo killall -15 nginx
sudo pkill -f "python worker.py"
```

Init Evolution: SysVinit to `systemd`

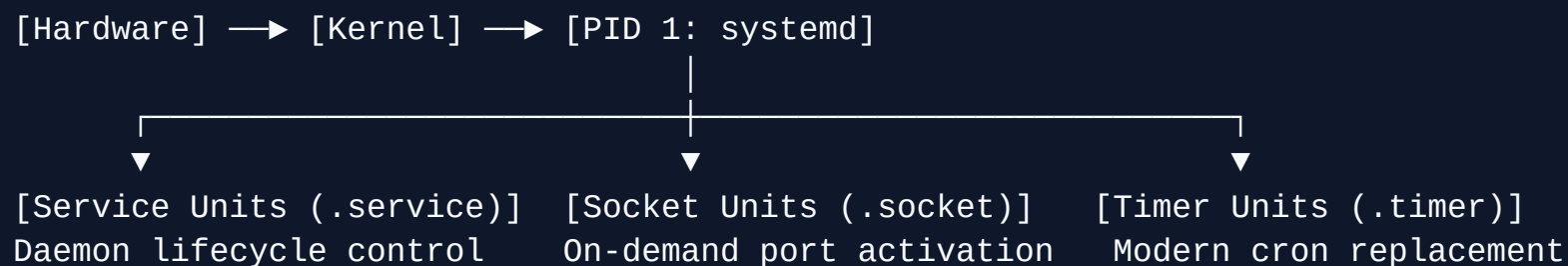
Modern enterprise Linux distributions have adopted `systemd` as the unified initialization system and service manager:

Legacy SysVinit

- Sequential boot: Services started one by one via shell scripts in `/etc/init.d/`.
- Slow boot performance.
- No built-in supervision: Crashed daemons remained dead.
- Child processes could escape tracking by double-forking.

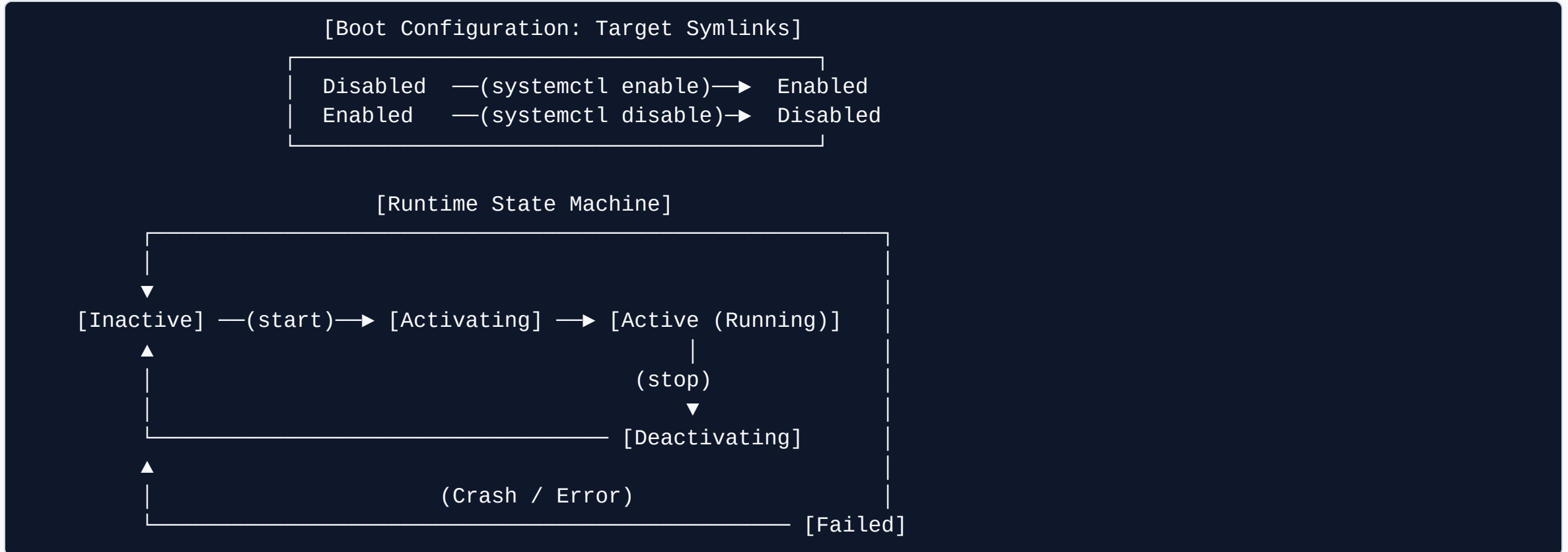
Modern `systemd`

- Concurrent, parallelized boot activation.
- Uses Linux **Control Groups (cgroups)** to track every child process cleanly.
- Automatic restart policies (`Restart=on-failure`).
- Unified operational syntax across Ubuntu, Rocky, Debian, RHEL, and SUSE.



⚙️ Systemd Service Lifecycle & State Machine

Systemd decouples **Runtime Execution** from **Boot Configuration**:



Systemd Management Commands (systemctl)

Operational Task	Command Syntax	Description
Inspect detailed status	<code>systemctl status nginx</code>	Active state, PID, memory usage, cgroup, and recent log lines
Start service immediately	<code>sudo systemctl start nginx</code>	Transitions service from Inactive to Active
Stop running service	<code>sudo systemctl stop nginx</code>	Sends SIGTERM, followed by SIGKILL if timeout expires
Restart service	<code>sudo systemctl restart nginx</code>	Full teardown and re-execution
Reload configuration	<code>sudo systemctl reload nginx</code>	Sends SIGHUP to reload configs without connection drops
Enable on system boot	<code>sudo systemctl enable nginx</code>	Creates target symlink for boot activation
Disable on system boot	<code>sudo systemctl disable nginx</code>	Removes target boot symlink
Enable AND start immediately	<code>sudo systemctl enable --now nginx</code>	Combines <code>enable</code> and <code>start</code>
Check boot enablement	<code>systemctl is-enabled nginx</code>	Returns <code>enabled</code> or <code>disabled</code>
Check runtime state	<code>systemctl is-active nginx</code>	Returns <code>active</code> or <code>inactive</code>

[!TIP]
Hands-On Lab 5 Milestone: Start Nginx, test port responses via `curl`, inspect service status, and verify autostart enablement.

DAY 2: Services, Networking, Security & GPU Admin

Module 6: Networking & SSH in Multipass

TCP/IP Sockets • Port Binding • Asymmetric Cryptography • Ed25519 • Cross-VM SSH

Linux Networking Essentials & Port Binding

Linux provides a kernel-managed network stack configured through user-space tools:

Network Interfaces

- **Loopback (`lo` / `127.0.0.1`):** Internal interface for local inter-process communication.
- **Physical & Virtual NICs:**
 - `eth0` or persistent names like `enp0s3` .
 - In Multipass: Assigned private IP addresses on an isolated bridge subnet (`10.x.x.x`).

Network Ports & Sockets

A **socket** is an IP address + Port number + Protocol binding (e.g. `0.0.0.0:80/TCP`).

- **Standard Ports:**
 - `22` : SSH (Secure Shell)
 - `80` : HTTP Web Traffic
 - `443` : HTTPS Encrypted Traffic
- **Privileged Ports (0–1023):** Require `root/CAP_NET_BIND_SERVICE` capabilities to bind.

Network diagnostic commands:

```
ip addr show          # View assigned IP addresses (replaces deprecated ifconfig)
ip route show         # View routing table and default gateway
ss -tulpn             # Inspect active listening sockets and owning PIDs
```

Asymmetric Cryptography & The SSH Architecture

SSH (*Secure Shell*, port 22) uses **Public-Key (Asymmetric) Cryptography** to secure remote connections:

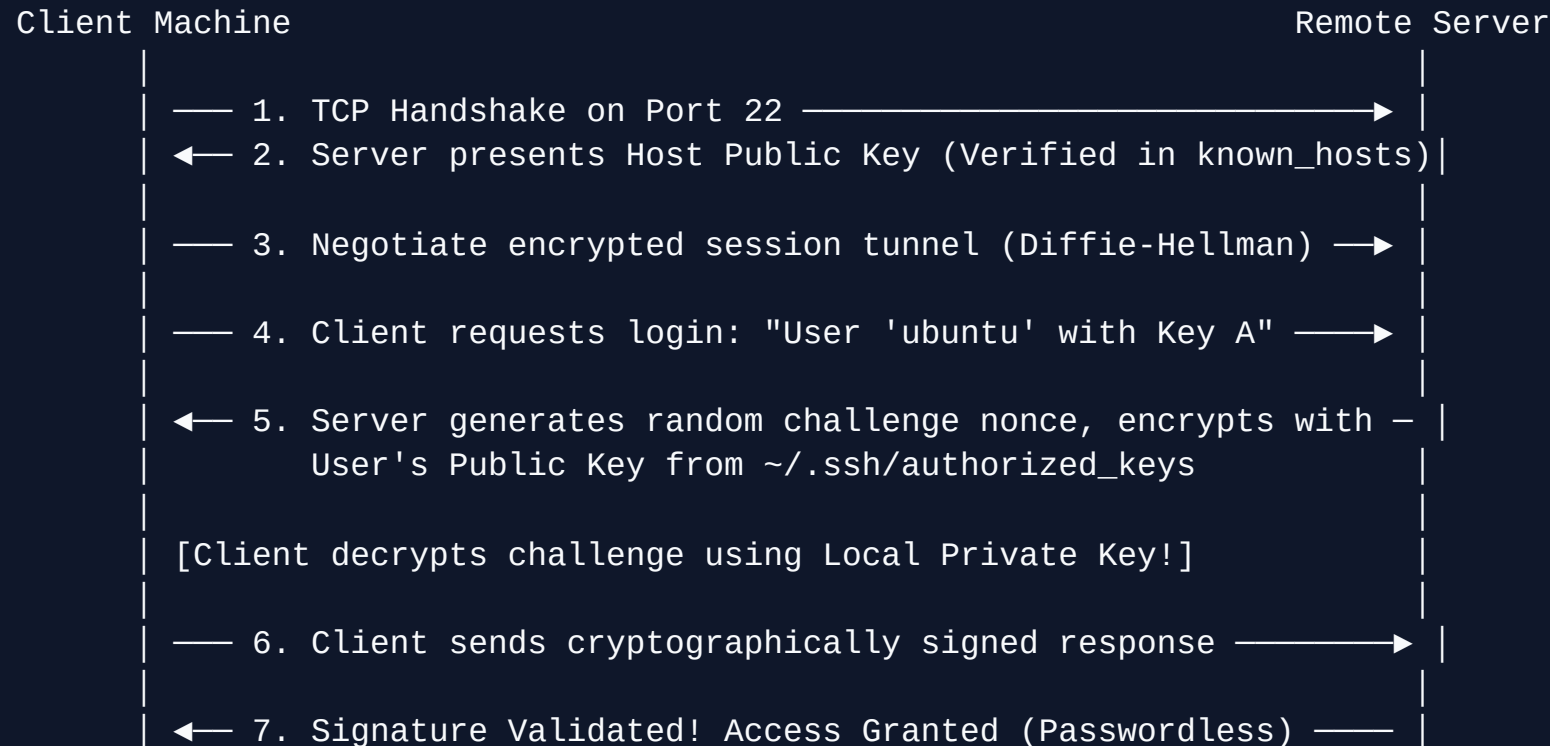


[!IMPORTANT]

Core Principle: Your **Private Key** must NEVER leave your client machine! The remote server only requires your **Public Key** placed in `~/.ssh/authorized_keys`.

SSH Authentication Handshake: Under the Hood

How does a server verify client identity without ever seeing the private key?



Key Algorithms: Why Ed25519 Over RSA?

Modern enterprise infrastructure specifies **Ed25519** (Edwards-curve Digital Signature Algorithm) for SSH authentication:

Ed25519 (Recommended Standard)

- Uses Curve25519 elliptic curve mathematics.
- Compact 256-bit key length provides security equivalent to a 3072-bit RSA key.
- Resistant to side-channel and timing attacks.
- Shorter key string: easier to manage in configuration files and CI/CD pipelines.
- Fast key generation and verification:

```
ssh-keygen -t ed25519 -N "" -f ~/.ssh/id_ed25519
```

RSA (Legacy Standard)

- Relies on integer factorization of large prime numbers.
- Requires at least 3072 or 4096 bits today.
- Larger key payloads; slower computation.
- Older SHA-1 based RSA signatures are disabled by default in modern OpenSSH releases.

[!CAUTION]

Strict Permission Requirements: OpenSSH rejects connection attempts if permissions are too open. Ensure `chmod 700 ~/.ssh` and `chmod 600 ~/.ssh/id_ed25519`.

[!TIP]

Hands-On Lab 6 Milestone: Generate an Ed25519 keypair, configure passwordless login to `localhost`, and SSH between the Ubuntu and Rocky Linux VMs.

LUNCH BREAK (13:00 – 14:00)

Mid-Day Break & Afternoon Preview

Take an hour to recharge and refresh.

Afternoon Sessions:

- **Module 7 (14:00 – 15:00):** Security Essentials & System Logs (`journalctl` , `tail -f` , `grep`)
- **Module 8 (15:00 – 16:00):** NVIDIA GPU Management & Multi-Instance GPU (MIG) on Remote GPU Server!

DAY 2: Services, Networking, Security & GPU Admin

Module 7: Security Essentials & System Logs

Logging Architecture • Syslog vs Journald • Binary Journals • Querying with journalctl • grep

Linux Logging Architecture: Syslog vs Journald

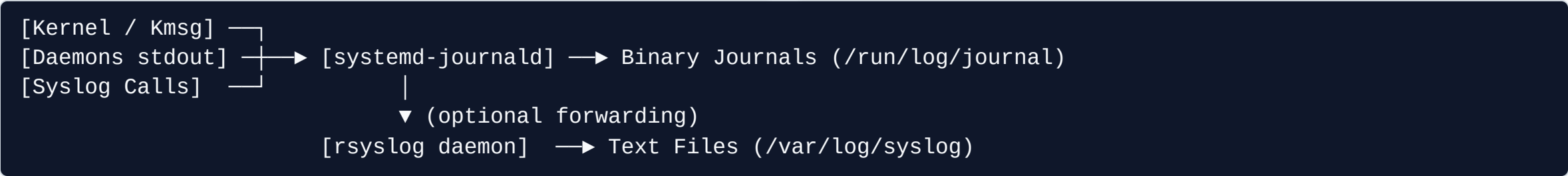
System logs provide essential observability for service debugging, auditing, and incident response:

Traditional Syslog (rsyslog)

- Appends plaintext lines to files in `/var/log/`.
- Standard log destinations:
 - Ubuntu: `/var/log/syslog` , `/var/log/auth.log`
 - Rocky Linux: `/var/log/messages` , `/var/log/secure`
- Managed by `logrotate` to prevent disk filling.
- Queried with standard text tools: `cat` , `grep` , `tail` .

`systemd-journald` (Modern Unified Log)

- Centralized binary ring-buffer logging daemon.
- Aggregates kernel messages, boot events, syslog calls, and service `stdout` / `stderr` .
- Enriches every record with structured metadata (PID, UID, Systemd Unit).
- Fast indexed searches and tamper resistance.
- Queried using the `journalctl` utility.





Structured Log Queries with `journalctl`

Structured journal metadata allows targeted queries without requiring complex pipeline filtering:

Common Query Filters

```
# 1. Filter by specific systemd service:
sudo journalctl -u nginx

# 2. Live stream new log entries:
sudo journalctl -u nginx -f

# 3. View logs from current system boot:
sudo journalctl -b

# 4. View logs from previous boot:
sudo journalctl -b -1

# 5. Filter by time window:
sudo journalctl --since "1 hour ago"
sudo journalctl --since "2026-09-07 10:00:00"
```

Severity Priority Filtering (`-p`)

Journal events follow standard RFC 5424 severity codes:

- 0 : emerg (System unusable)
- 1 : alert (Immediate action required)
- 2 : crit (Critical system conditions)
- 3 : **err (Error conditions)**
- 4 : warning (Warning conditions)
- 5 : notice (Normal but significant)
- 6 : info (Informational)
- 7 : debug (Debug messages)

```
# Display errors and critical faults from current boot:
sudo journalctl -p err -b
```

Real-Time Monitoring & Pipeline Filtering

Sysadmins combine live log following with pattern matching for active incident troubleshooting:

Live File Following (`tail -f`)

Monitors growing log files and streams newly appended lines to the screen:

```
# Follow Nginx HTTP access requests:
sudo tail -f /var/log/nginx/access.log

# Follow Nginx application error logs:
sudo tail -f /var/log/nginx/error.log
```

Press `ctrl + c` to stop following.

Pattern Filtering with `grep`

Searches text streams for matching regular expressions:

- `grep -i "error"` : Case-insensitive matching.
- `grep -n "fail"` : Output line numbers.
- `grep -v "info"` : Invert match (exclude lines).
- `grep -E "404|500"` : Extended regex matching.

```
# Filter for failed SSH login attempts:
sudo grep "Failed password" /var/log/auth.log
```

[!TIP]

Hands-On Lab 7 Milestone: Generate simulated HTTP traffic to Nginx, inspect live logs with `journalctl -f` and `tail -f`, and filter security events using `grep`.

DAY 2: Services, Networking, Security & GPU Admin

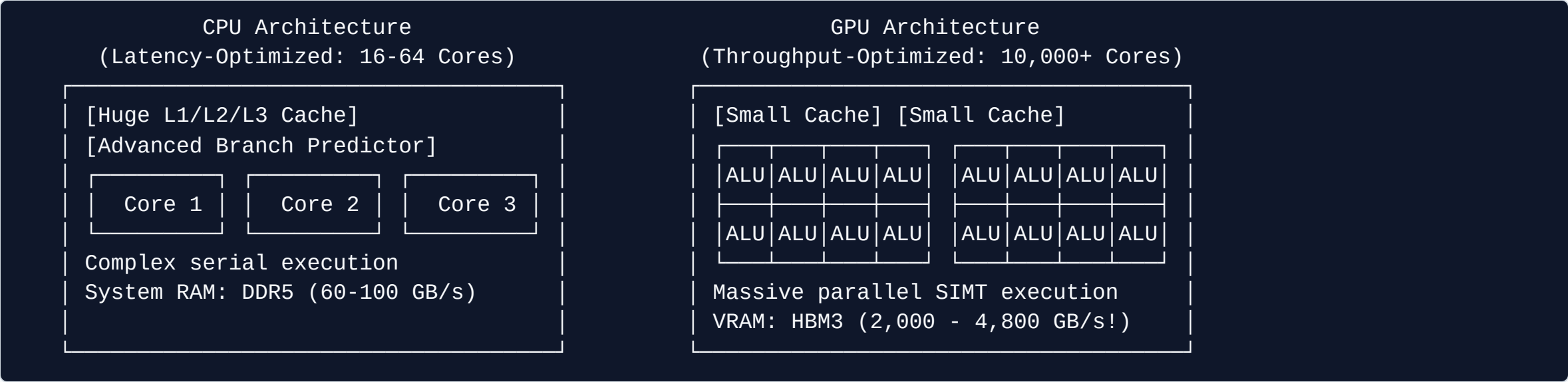
Module 8: NVIDIA GPU Management & MIG

GPU Architecture • Software Stack • nvidia-smi Telemetry • Multi-Instance GPU (MIG)

Environment: Dedicated Remote Server with Enterprise NVIDIA GPUs (A100 / H200)

CPU vs GPU Compute Paradigms in Modern AI

Modern AI model training and LLM inference require massive parallel computation:

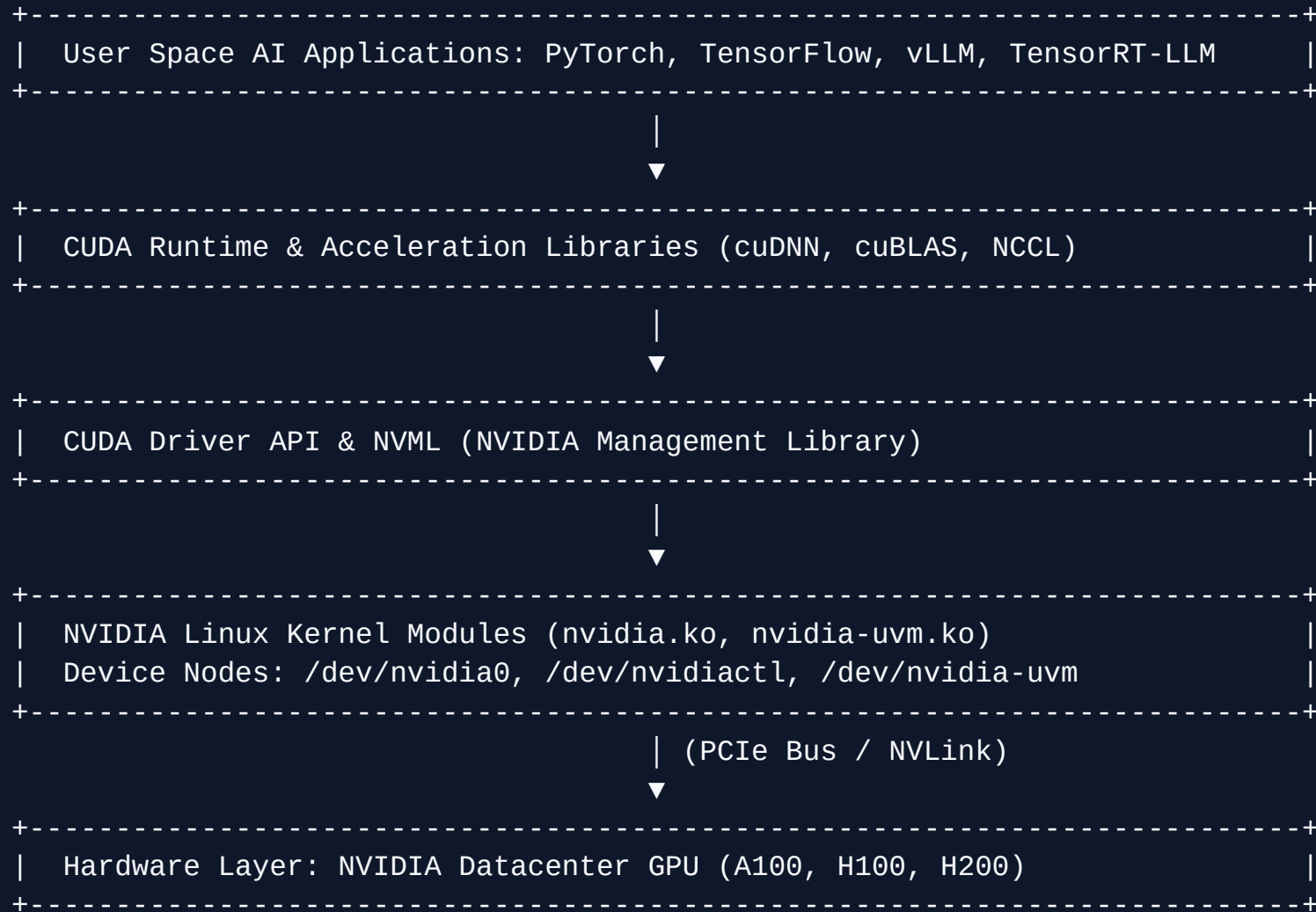


- **CPU:** Optimized for sequential execution, complex control logic, and system orchestration.
- **GPU (SIMT):** Optimized for high-throughput parallel matrix multiplication and tensor arithmetic.



The NVIDIA Linux Software Stack

Accessing NVIDIA GPU hardware from Linux involves several coordinated software layers:





Real-Time GPU Telemetry: `nvidia-smi` Essentials

The **NVIDIA System Management Interface** (`nvidia-smi`) monitors GPU operational status:

```
+-----+
| NVIDIA-SMI 550.54.15                Driver Version: 550.54.15          CUDA Version: 12.4          |
+-----+-----+-----+-----+-----+-----+
| GPU  Name                Persistence-M | Bus-Id        Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |      Memory-Usage | GPU-Util  Compute M. |
|                                           | MIG M.         |
+-----+-----+-----+-----+-----+-----+
|   0   NVIDIA A100-SXM4-80GB         On   | 00000000:00:04.0 Off |              0      |
| N/A   38C    P0              68W / 400W | 1420MiB / 81920MiB |      5%      Default |
|                                           |                  Enabled |
+-----+-----+-----+-----+-----+-----+
| Processes:                                     |
|  GPU   GI    CI          PID    Type   Process name                      GPU Memory |
|        ID    ID                                   Usage      |
+-----+-----+-----+-----+-----+-----+
|    0   --    --        14298     C   python3 /workspace/train.py        1410MiB |
+-----+-----+-----+-----+-----+-----+
```

- **Persistence Mode:** Ensures driver stays loaded in kernel memory (`Persistence-M: On`).
- **Power Usage vs Capacity:** Tracks current draw against TDP limit (e.g. 68W / 400W).
- **Process Types:** `c` = Compute / CUDA application, `G` = Graphics application.



Telemetry Automation & Process Tracking

`nvidia-smi` supports scriptable querying for monitoring pipelines, metrics exporters, and alerting:



Live Terminal Monitor

Continuously refresh GPU status every second:

```
nvidia-smi -l 1
```



Scriptable CSV Telemetry Query

Extract structured metrics for monitoring agents:

```
nvidia-smi --query-gpu=index,name,temperature.gpu,utilization.gpu,memory.used,memory.total --format=csv
```

Example CSV output:

```
index, name, temperature.gpu, utilization.gpu [%], memory.used [MiB], memory.total [MiB]
0, NVIDIA A100-SXM4-80GB, 38, 5 %, 1420 MiB, 81920 MiB
```



Process Tracking

Identify user PIDs consuming GPU VRAM:

```
nvidia-smi --query-compute-apps=gpu_uuid,pid,process_name,used_gpu_memory --format=csv
```



Critical Operational Checks

- **Thermal Limits:** Check for temperatures exceeding 80°C.
- **Out of Memory (OOM):** Track if allocated memory nears capacity.
- **Orphan Processes:** Locate abandoned Python processes holding VRAM after jobs terminate.

The Multi-Tenant Problem in AI Clusters

Enterprise GPUs (e.g. A100-80GB, H200-141GB) represent significant infrastructure investments:

Software Time-Slicing Limitations

- Multiple jobs share a single GPU via software scheduling.
- **The Noisy Neighbor Issue:**
 - One application with an unoptimized memory leak causes a CUDA OOM crash across all concurrent jobs!
 - Unpredictable execution latency.
 - Zero memory hardware isolation.

Enterprise Multi-Tenant Requirements

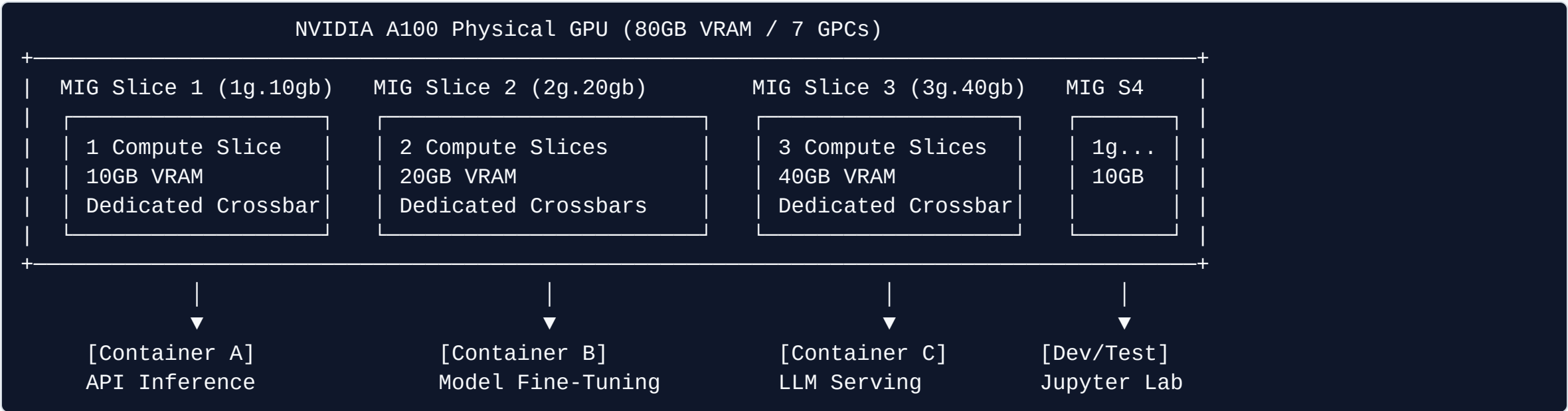
1. **Guaranteed Quality of Service (QoS):** Deterministic latency for real-time inference.
2. **Hard Fault Isolation:** A crash in one workload never affects other workloads.
3. **Hardware Memory Partitioning:** Dedicated, non-overlapping VRAM segments.
4. **Dedicated Compute Paths:** Isolated streaming multiprocessors and cache lines.

[!IMPORTANT]

The Enterprise Solution: NVIDIA **Multi-Instance GPU (MIG)** silicon partitioning technology.

Multi-Instance GPU (MIG): Silicon-Level Partitioning

Available on NVIDIA Ampere (A100) and Hopper (H100/H200) architectures, MIG partitions a single physical GPU into up to **7 hardware-isolated instances**:



- Each instance has dedicated **Streaming Multiprocessors (SMs)**, **Memory Controllers**, and **L2 Cache**.
- Hardware fault isolation guarantees that a crash in one slice does not impact any other slice!



MIG Architecture: GI vs CI & Profile Sizing

MIG defines two hierarchical layers of partitioning:

1. GPU Instance (GI)

- Primary hardware partition.
- Allocates memory controllers, VRAM capacity, and compute slices.

2. Compute Instance (CI)

- Execution slice within a GPU Instance.
- The `-c` flag creates matching 1:1 Compute Instances automatically.



Profile Naming Scheme

{Compute Slices}g. {Memory in GB}gb

- `1g.10gb` : 1 compute slice, 10GB VRAM (Up to 7 instances)
 - Ideal for: Web inference APIs, embeddings, dev testing.
- `2g.20gb` : 2 compute slices, 20GB VRAM (Up to 3 instances)
 - Ideal for: Computer vision, small LLM inference.
- `3g.40gb` : 3 compute slices, 40GB VRAM (Up to 2 instances)
 - Ideal for: Medium model fine-tuning (LoRA), data pipelines.
- `7g.80gb` : Full GPU resources under MIG isolation.



The 4-Phase MIG Operational Lifecycle

Phase 1: Enable MIG Mode → Phase 2: Create Partitions → Phase 3: Bind Workload → Phase 4: Teardown
 (sudo nvidia-smi -mig 1) (nvidia-smi mig -cgi ...) (CUDA_VISIBLE_DEVICES) (nvidia-smi mig -dgi)

Phase 1: Mode Enablement

Requires root privileges:

```
sudo nvidia-smi -i 0 -mig 1
```

Phase 2: Profile Selection & Slicing

List supported profile IDs, then create partitions:

```
# List supported profiles:
nvidia-smi mig -lgip -i 0

# Create a 20GB slice and a 10GB slice:
sudo nvidia-smi mig -cgi 14,19 -C
```

Phase 3: Workload Binding

Find created instance UUIDs:

```
nvidia-smi mig -lgi
```

Bind target application to that slice only:

```
export CUDA_VISIBLE_DEVICES="MIG-<UUID>"
python3 inference_server.py
```

Phase 4: Teardown & Reset

```
# Destroy created MIG instances:
sudo nvidia-smi mig -dgi -i 0

# Disable MIG mode:
sudo nvidia-smi -i 0 -mig 0
```

The 4-Phase MIG Operational Lifecycle (cont.)

[!TIP]

Hands-On Lab 8 Milestone: SSH into the remote dedicated GPU node, query telemetry with `nvidia-smi`, enable MIG mode, carve GPU instances, bind a workload, and perform clean teardown.

Using NVIDIA MIG Parted to create MIG partitions

What is NVIDIA MIG Partition Editor (MIG Parted)

MIG (short for **Multi-Instance GPU**) is a software that allows NVIDIA GPU to be sliced into mini GPUs with a fixed partition of memory and a fixed partition of compute resources.

Why MIG Parted?

1. Less command to run compared to using just `nvidia-smi`.
2. The configuration can be saved in git repository for backup and sharing purposes.
3. Can be easily paired with `systemd` to reapply the profile after each reboot.

How to use MIG Parted to create MIG partitions?

1. [Download](#) and install nvidia-mig-parted.
2. List supported MIG profiles for the GPU card

```
sudo nvidia-smi mig -lgip
```

3. Create a configuration file in /home/apps/nvidia-mig-parted/config.yaml. Get a sample from [here](#).
4. Apply the desired profiles. For example, to activate a profile named all-balanced, use below command

```
sudo mig-parted -f /home/apps/nvidia-mig-parted/config.yaml -c all-balanced
```

5. Verify that the MIG profile has been applied

```
sudo nvidia-smi mig -lgi
```

Automating MIG profiling on reboot

By default, mig-parted would not survive reboot. To reapply MIG parted on each reboot:

1. Create a systemd service:

```
sudo systemctl edit --full --force nvidia-mig-parted.service
```

2. Fill up the file with content from [nvidia-mig-parted.service](#).

3. Save and exit.

4. Reload systemd daemon:

```
sudo systemctl daemon-reload
```

5. Start the service, and enable it to start on boot:

```
sudo systemctl enable --now nvidia-mig-parted
```

6. Verify that the service is now started, and the MIG profile has been activated:

```
sudo systemctl status nvidia-mig-parted; sudo nvidia-smi mig -lgi
```

Course Wrap-up & Enterprise Next Steps

Core Competencies Mastered

- **Day 1: Foundations & Storage**
 - Terminal navigation, pipes, stream redirection, nano.
 - Multi-user security, sudo, and POSIX permissions (`chmod 755/644/600`).
 - Dual-ecosystem package management (`apt` vs `dnf`).
 - Storage telemetry (`df` / `du`), loopback disks, ext4 vs XFS.
- **Day 2: Operations, Security & HPC**
 - Process lifecycle, signals (`SIGTERM` / `SIGKILL`), and systemd units.
 - Network sockets, port binding, and Ed25519 SSH keys.
 - Logging observability with `systemd-journald` and `grep` .
 - Datacenter NVIDIA GPU monitoring and MIG hardware partitioning.

? Questions & Discussion

Practical Linux System Administration Training

Participant Lab Guide: `LINUX_TRAINEE_WORKBOOK.md`

Course Syllabus & Timetable: `timetable.md` / `README.md`

Slide Deck: `SLIDES.md`

Repository materials dedicated under CC0 1.0 Universal Public Domain.